

# Procedural Stress Testing of Actuarial Models: Design and Preliminary Evaluation of the ARCCRAFT Framework

*Does integrating explicit multi-period transitions, structural validation, separated random streams, replay semantics, and failure attribution produce measurable improvements in reproducibility, diagnostic information, failure detection, or empirical prediction relative to transparent comparators, and at what cost?*

## Abstract

Insurance stress tests commonly apply predefined shocks or simulate losses from static frequency-severity models. These approaches estimate adverse outcomes but do not necessarily integrate scenario sequencing, structural consistency, deterministic replay, and failure attribution. This paper proposes ARCCRAFT (Actuarial Risk & Contingency Computational, Reproducible Architecture for Failure Testing), a versioned procedural architecture in which synthetic insurance worlds evolve over multiple periods under explicit states, transitions, bounds, stochastic streams, and failure predicates. The evaluation separates computational verification from empirical prediction. First, a retrospective motor-insurance study uses 354,140 policy-year observations from 2022-2024 and two temporal forecast origins. Policy-level Poisson pseudo-maximum-likelihood frequency modelling reduces held-out count deviance from 0.8481 to 0.7871 in 2023 and from 0.9153 to 0.8330 in 2024, but does not improve every aggregate error. A policy-level cost model is unstable across years: its incurred-loss error is +9.58% in 2023 and -4.44% in 2024. No model covers observed incurred loss with its process-only interval, and no model covers the 2024 claim count. Second, a synthetic ablation study using three seeds and 10,000 worlds per configuration establishes exact local replay and shows that named PCG64 substreams isolate claim and behavioural results from irrelevant world-stream consumption. Validator and transition ablations alter the selected population and coded failure rates, but the full model's failures are almost entirely generated by its operational traceability gate. The results support ARCCRAFT as a reproducible audit and diagnostic architecture in the inspected environment; they do not establish calibrated scenario plausibility, general predictive superiority, causal failure attribution, or external validity. Independent portfolios, claim-level losses, operational labels, and prospectively untouched forecast periods are required for confirmatory validation.

**Keywords:** actuarial stress testing; procedural simulation; non-life insurance; model risk; reproducibility; reverse stress testing; predictive calibration.

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## 1. Introduction

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Actuarial decisions under stress require more than an estimate of expected loss. Insurers and regulators also need to know how adverse conditions interact over time, which combinations cross operational or solvency thresholds, whether a critical result can be replayed, and which assumptions or events are associated with failure. Conventional frequency-severity models and Monte Carlo simulations remain essential for loss projection, but the surrounding workflow often treats scenario construction, validation, random-number governance, and diagnostic attribution as separate activities.

This separation creates several risks. A simulated outcome can be numerically extreme but structurally inconsistent. A scenario can be labelled severe and plausible without a quantitative plausibility criterion. A result can be reproducible under one execution order yet change after an unrelated component consumes random numbers. A compact point error can coexist with poor predictive-interval coverage. Finally, a failure record can identify the threshold that was crossed without establishing the real-world cause of that failure. These properties (loss projection, scenario plausibility, computational reproducibility, predictive calibration, and attribution) must therefore be evaluated separately.

ARCCRAFT is proposed as an integration architecture for this problem. It represents a synthetic environment as a versioned, time-dependent world; allocates independent pseudorandom streams to world generation, claims, and behaviour; validates initial states through explicit pass, repair, or reject decisions; simu-

lates multi-period outcomes; and records failed worlds in a seed-addressable Failure Atlas. The framework also includes a local reverse-stress search. These components build on established work rather than constituting a new actuarial theory.

The research question is: **Does integrating explicit multi-period transitions, structural validation, separated random streams, replay semantics, and failure attribution produce measurable improvements in reproducibility, diagnostic information, failure detection, or empirical prediction relative to transparent comparators, and at what cost?** The evaluation is deliberately falsifiable. ARCCRAFT may add governance value without improving predictive accuracy, and individual architectural components may have little effect under the current grammar.

This study makes four bounded contributions. First, it specifies the implemented architecture and maps its claims to executable functions and versioned outputs. Second, it audits a complete motor-insurance table and implements matched aggregate and policy-level actuarial comparators over two temporal origins. Third, it tests local deterministic replay, stream isolation, validator policy, state transitions, and runtime through controlled ablations. Fourth, it reports negative findings, including interval misses, decile miscalibration, lack of policy-severity superiority, and dominance of the traceability gate, rather than defining validation from favourable point estimates.

## 2. Literature review and hypotheses

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## 2.1. Actuarial loss modelling and the limits of static projection

Non-life insurance models commonly decompose aggregate loss into claim frequency and claim severity. This decomposition is attractive because claim counts contain many zeros, positive claim amounts are typically right-skewed, and the two components may respond differently to rating and environmental variables. In its simplest form, a collective-risk model generates a claim count and then draws claim amounts, often under conditional independence. Such models remain useful transparent baselines, but the independence assumption is not innocuous when the number and average size of claims share unobserved risk factors or evolve jointly over time.

Several studies have relaxed this assumption. Garrido et al. [1] incorporate dependence by including claim count in a conditional severity model and show how the resulting pure premium differs from the independent frequency-severity formulation. Shi et al. [2] develop hurdle and copula constructions for policy-level frequency and severity, while Frees et al. [3] extend the problem to multivariate insurance outcomes. Lee and Shi [4] add longitudinal dependence, allowing past claim frequency and severity to inform the future predictive distribution. Jeong and Valdez [5] similarly use dependent random effects to construct predictive compound-risk models, and Gao and Li [6] link waiting time and average severity to obtain a predictive aggregate-loss distribution. Together, these studies establish that a credible benchmark for a multi-period actuarial framework should not automatically assume that frequency, severity and their temporal dynamics are independent.

The implications for stress testing are twofold.

First, a joint frequency-and-severity shock is not equivalent to two unrelated marginal multipliers when the components share latent or observed dependencies. Second, a multi-period stress environment cannot be evaluated adequately by comparing only its terminal expected loss with a historical mean. The temporal ordering of shocks, transitions and behavioural responses may affect both the path to failure and the distribution of the terminal outcome. A procedural stress architecture must therefore demonstrate what its transition rules add beyond a transparent longitudinal frequency-severity model rather than treating temporal sequencing as intrinsically superior.

Flexible actuarial models provide additional comparators. Klein et al. [7] use Bayesian generalized additive models for location, scale and shape to let covariates affect more than the conditional mean. Modern tree and boosting methods can capture nonlinearities and interactions, but their predictive gains must be balanced against interpretability and calibration. These methods are relevant to ARCCRAFT because a procedural environment could appear useful simply because it embeds nonlinear dependence that a weak baseline omits. Comparator selection must therefore include at least one transparent conventional model and, where the data support it, one dependence-aware or flexible alternative.

## 2.2. From predefined shocks to systematic stress scenarios

Insurance stress testing is intended to examine resilience under adverse conditions, not merely to reproduce the central predictive distribution. EIOPA [8] frames insurance stress testing around clearly specified objectives, material risks and

severe-but-plausible scenarios. This supervisory framing makes scenario design a substantive modelling decision: an adverse outcome is informative only if the scenario producing it is coherent with the exercise’s purpose and its plausibility criterion. A large loss generated by an internally inconsistent or unconstrained combination of variables is not automatically a credible stress-test result.

The scenario-selection literature formalizes this problem. Breuer et al. [9] characterize the tension between severity and plausibility and propose searching systematically for damaging scenarios inside an admissible region. Breuer and Csiszar [10] use entropic plausibility constraints to construct systematic stress tests, while Glasserman et al. [11] use empirical likelihood to identify plausible combinations of factors associated with extreme losses. Flood and Korenko [12] position systematic scenario selection between hand-picked shocks and reverse stress testing, emphasizing internally consistent multidimensional shocks and the reduction of blind spots.

These approaches reveal an important distinction for ARCCRAFT. A structural validator can reject impossible combinations or repair violated bounds, but internal consistency is not identical to statistical or economic plausibility. A world can satisfy coded invariants while remaining implausible relative to observed joint dynamics. Conversely, a repair operation may create a formally admissible world while altering the scenario distribution in ways that bias tail estimates. The validator must therefore be evaluated as part of the data-generating process. Its acceptance, repair and rejection rates, and the resulting changes in marginal and joint output distributions, are empirical outcomes rather than implementation details.

The literature also cautions against presenting a fixed stress label as self-validating. Terms such as *adverse* and *severe plausible* need quantitative definitions: probability regions, divergence constraints, historical or expert bounds, or another documented criterion. ARCCRAFT’s proposed regime taxonomy can organize experiments, but the taxonomy contributes scientifically only when each regime is operationalized and its relationship to the baseline distribution is reported.

### 2.3. Reverse stress testing and failure attribution

Forward stress testing begins with an adverse scenario and measures its consequences. Reverse stress testing begins with a failure condition and asks which changes to inputs or model distributions can produce that condition. Pesenti et al. [13] provide a particularly relevant insurance application. Their reverse sensitivity framework specifies an output stress, derives a probability measure that satisfies the stress while minimizing divergence from the baseline, and evaluates how the distributions of input factors change. This design produces an interpretable ranking of factors associated with “breaking” a model without requiring repeated evaluation of an expensive aggregation function.

Kroell et al. [14] extend reverse stress concepts to dynamic loss models. They construct stressed probability measures for a compound Poisson process over time under terminal constraints, thereby showing that dynamic reverse stress testing is already an established research direction. ARCCRAFT should therefore not claim novelty for dynamic or reverse stress testing themselves. The relevant question is narrower: whether its

procedural representation and Failure Atlas provide an auditable, useful link between simulated trajectories, first failure times, threshold conditions and replayable seeds that is not already supplied by the comparator methods.

Sensitivity analysis provides further context. Morris [15] offers an efficient screening design for identifying influential inputs in computational experiments, whereas variance-based methods estimate main and interaction effects across the input space Saltelli et al. [16, 17]. These methods measure influence under specified distributions; they do not by themselves establish causality. A Failure Atlas that associates a failed seed with prominent variables should consequently use terms such as *attribution*, *association* or *influence* unless it is tested against implanted causal mechanisms or another identification design.

This requirement suggests a falsifiable experiment. Synthetic worlds can be generated with registered faults whose responsible transition, dependency or event mechanism is known. ARCCRAFT and comparator diagnostics can then be evaluated blindly on their ability to detect the failure and recover its responsible mechanism. Precision, recall, false-discovery rate, detection delay and attribution rank would provide stronger evidence than selected examples of apparently informative failure traces.

## 2.4. Synthetic insurance environments and open benchmarks

Synthetic insurance data are not new. Gabrielli and Wuthrich [18] present an individual claims-history simulation machine calibrated on extensive real claim histories. Avanzi et al. [19] introduce SynthETIC, an open simulator that gives users control over claim occurrence, notification,

payment timing, inflation, dependencies and other realistic features. Avanzi et al. [20] extend this ecosystem through SPLICE, which simulates paid and incurred claim histories and revisions of case estimates. These systems demonstrate that control over an insurance data-generating mechanism is valuable for benchmarking methods when confidential individual-claim data are unavailable.

Open artificial insurers provide an even closer architectural precedent. Wolf et al. [21] introduce openIRM as a publicly accessible internal risk model of an artificial life insurer for benchmarking nested-simulation and other actuarial methods under Solvency II. It integrates an economic scenario generator, asset-liability dynamics and capital estimation, and is calibrated over multiple market dates. This work shows that openness, time-dependent risk factors and reproducible benchmarking are already active contributions in actuarial science.

ARCCRAFT must consequently be distinguished by a specific combination of functions rather than by synthetic simulation alone. Its provisional distinction is the integration of versioned world specifications, explicit state transitions and events, separate stochastic streams, structural accept-repair-reject decisions, deterministic seed-level replay and a persistent failure-attribution record. Whether this integration yields scientific value remains an empirical question. It could improve diagnosis and governance without improving predictive accuracy; it could also increase computational cost or induce selection bias through validator decisions. The evaluation must allow each of these outcomes.

Synthetic evidence also has a fundamental limitation. A simulator can confirm that an anal-

ysis procedure recovers patterns deliberately embedded in the simulator, but it cannot independently demonstrate that those patterns describe the real insurance system. External portfolio evidence is therefore necessary. Synthetic fault injection can test failure detection and attribution, while independent observed outcomes must test predictive calibration and practical relevance. Neither form of evidence substitutes for the other.

## 2.5. Reproducibility is not validation

Computational reproducibility requires enough information and artifacts to regenerate reported results. Peng [22] and Stodden [23] treat accessible code, data and workflow information as central to reproducible computational research. Stodden et al. [24] further emphasize institutional practices that make computational methods inspectable and repeatable. In stochastic simulation, controlled random streams and substreams support designed comparisons and component isolation [25]. PCG64 provides an implementation choice for pseudo-random generation, but reporting a generator name and seed is insufficient unless software versions, stream construction and replay tolerances are also recorded.

Reproducibility nevertheless answers only whether an experiment can be repeated, not whether its model is empirically adequate. Sargent [26] distinguishes conceptual-model validity, model verification, operational validity and data validity. Kleijnen and Sargent [27] likewise distinguish fitting from validation and tie the required accuracy to the model's intended purpose. In actuarial practice, ASOP No. 56 requires attention to intended use, input reconciliation, formula and logic checks, sensitivity testing, output validation, limitations and model risk [28].

ARCCRAFT should therefore report at least three separate claims. **Software verification** concerns whether the coded engine implements its stated rules and reproduces registered runs. **Empirical validation** concerns whether output distributions are compatible with independent observations for their intended use. **Scenario plausibility** concerns whether adverse worlds remain defensible under empirical, regulatory or expert constraints. A successful replay establishes the first claim only. It cannot compensate for failed predictive coverage or an unjustified severe-plausible label.

The same separation applies to uncertainty. Der Kiureghian and Ditlevsen [29] distinguish aleatory variability from epistemic uncertainty, while Kennedy and O'Hagan [30] formalize the discrepancy between a computer model and the process it approximates. Barrieu and Scandolo [31] show how model risk can be quantified relative to alternative models. These perspectives imply that one stochastic stream should not silently represent process randomness, parameter uncertainty, behavioural uncertainty and model-form uncertainty as if they were interchangeable. ARCCRAFT's independent streams may improve experimental control, but the semantic meaning and calibration of each source of randomness must be explicit.

## 2.6. Predictive and tail evaluation

Point accuracy alone is inadequate for a stochastic stress framework. Gneiting and Raftery [32] show that proper scoring rules reward honest probabilistic forecasts, while Gneiting et al. [33] distinguish calibration from sharpness. Czado et al. [34] adapt predictive assessment to count data, which is directly relevant to insurance claim

counts. A model can achieve a small error in its predictive mean and still be poorly calibrated if the observed outcome falls systematically outside its prediction intervals.

Tail evaluation requires similar care. Value-at-Risk exceedances can be assessed through unconditional coverage [35] and conditional coverage [36], while joint evaluation of Value-at-Risk and Expected Shortfall requires methods consistent with their statistical properties [37]. These tests require repeated observations. A single annual outcome above a 97.5% predictive quantile is important negative evidence, but it is not enough to estimate a stable exceedance rate or to perform a well-powered backtest.

The present motor experiment illustrates why these distinctions matter. A retrospective projection calibrated on 2022-2023 produces a 2024 mean claim-count error of approximately -2.38% and an incurred-cost error of approximately -4.86%. Those central estimates appear close to the observed aggregates. However, the observed claim count, incurred cost and loss ratio all exceed their simulated 97.5% quantiles. The experiment therefore cannot support complete predictive validation. It indicates that the central level is reasonably close while the reported predictive distribution understates the realized upper-tail outcome. Because 2024 has already been inspected, further tuning against that year would convert it from evaluation evidence into development data.

Distribution shift is one possible explanation, but it must be tested rather than presumed. Cauchois et al. [38] show that predictive validity can deteriorate when training and evaluation distributions differ and develop prediction sets robust to bounded shifts. In the current applica-

tion, changes in exposure mix, inflation, reporting practice, dependence or unmodelled portfolio characteristics could each contribute. The empirical design should use rolling-origin evaluation, an independent portfolio or a later untouched period, and it should report both central and distributional performance.

## 2.7. Research gap and evaluation implications

The literature does not support describing ARCCRAFT as a new theory of actuarial stress testing. It supports a more precise gap. Established methods address dependent frequency-severity modelling, severe-plausible scenario selection, reverse stress testing, dynamic loss processes, synthetic claims, open artificial insurers, reproducible random streams, simulation validation and tail evaluation. What remains to be tested is whether integrating versioned procedural worlds, structural validation, controlled replay and seed-level failure attribution in one non-life workflow produces measurable benefits beyond those components and their conventional alternatives.

This gap leads directly to the research design. ARCCRAFT must be compared with deterministic projection, conventional frequency-severity Monte Carlo, a dependence-aware alternative and static stress scenarios under matched data and computational budgets. Its architectural components must be ablated. Validator repairs must be treated as distribution-changing interventions and evaluated accordingly. Failure detection and attribution must be tested against registered ground truth. Predictive performance must be assessed with proper scores, interval coverage and tail diagnostics, while reproducibility must be tested through exact or tolerance-based replay. Runtime, memory, rejection rates and ef-

fective accepted-sample size must be reported alongside any gain.

Under this design, ARCCRAFT can succeed in more than one defensible way. It may improve predictive distributions, detect failure configurations more efficiently, increase diagnostic accuracy, or strengthen reproducible model governance. It may also prove valuable mainly as an audit architecture without outperforming simpler predictive models. Conversely, if repairs distort the stress distribution, tail calibration remains poor, or the Failure Atlas does not recover known mechanisms, those findings would delimit the framework’s value. A credible contribution depends on reporting these possibilities rather than defining success after observing the results.

## 2.8. Testable expectations

The evaluation addresses the following expectations:

- **H1:** a registered seed reproduces the same worlds, metrics, and failure labels in the inspected environment;
- **H2:** named substreams isolate claims and behaviour from irrelevant consumption of the world stream;
- **H3:** the validator changes the accepted scenario population and reported failures;
- **H4:** multi-period state transitions change failure metrics relative to fixed validated states;
- **H5:** policy-level models improve held-out predictive performance relative to pooled means;
- **H6:** the Failure Atlas exposes replayable coded failure conditions, but causal diagnostic value

remains unsupported without external or implanted ground truth.

H5 is evaluated dimension by dimension rather than through a post-hoc composite score.

## 3. Research methodology

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### 3.1. Study design and evidence layers

The study has two evidence layers. The empirical layer evaluates aggregate and policy-level forecasts on a public motor portfolio. The procedural layer evaluates the internal behaviour of the synthetic ARCCRAFT engine. The layers are intentionally not conflated: the motor benchmark does not empirically validate the synthetic world grammar, and synthetic replay does not validate the motor predictions.

The protocol is retrospective. The 2024 outcome and an earlier implementation were inspected before the final policy-level specification was frozen. The final transformations, predictors, penalty, temporal origins, and metrics were recorded in `16_RETROSPECTIVE_ANALYSIS_SPECIFICATION.md` before executing the reported final notebook, but the analysis is not pre-registered or confirmatory.

### 3.2. Motor data and quality controls

The normalized SQLite table contains 354,140 policy-year records for 185,678 insured identifiers from 2022-2024 and 47 fields documented in the supplied dictionary. The database passed its integrity check, its schema exactly matched the dictionary, no duplicate insured-year keys or exact duplicate groups were detected, and premium,

claim-count, and incurred totals reconciled with their recorded components.

Quality controls covered completeness, range, categorical levels, internal consistency, temporal composition, and leakage. Missingness affected 513 vehicle values, two vehicle ages, and two licence values. One zero-exposure record had reported claims and was excluded from frequency fitting. Among 50,807 claim-bearing records, 8,543 (16.81%) had zero incurred cost; consequently, a positive-only Gamma severity model was not appropriate. The field `age_driving_licence` is described as a calendar year but ranges from 0 to 80, so it was retained only as an unidentified numeric proxy. `policy_status` was excluded because current cancellation status may contain post-inception information. The undocumented NA fuel category was retained as an explicit level. Portfolio rows increased from 67,172 in 2022 to 168,133 in 2024, and the largest business-type share moved by 40.57 percentage points, indicating material temporal composition change.

### 3.3. Empirical comparators

Two rolling temporal origins were used: 2022 to predict 2023, and 2022-2023 to predict 2024. Target-year policy characteristics and exposure were treated as known at forecast inception. Three specifications shared the same data partitions:

1. **B0, aggregate pooled:** pooled claims per unit exposure and pooled incurred loss per reported claim;
2. **B1, policy frequency with pooled severity:** a policy-level frequency model and pooled cost per claim;

3. **B2, policy frequency with policy severity:** policy-level models for both frequency and incurred cost per reported claim.

For policy (i), frequency was fitted by penalized Poisson pseudo-maximum likelihood,

$$\mu_i^N = E_i \exp(x_i^\top \beta), \quad (3.1)$$

where  $(E_i > 0)$  is exposure. The cost model used total incurred amount as the response and reported claim count as exposure,

$$\mu_i^L = N_i \exp(x_i^\top \gamma), \quad (3.2)$$

on claim-bearing calibration rows. This is a non-negative conditional-mean model and not a literal claim-level Poisson distribution for currency. It accommodates zero-incurred claim rows but cannot recover individual loss severity from policy-year aggregates.

The design matrix included driver age, vehicle age, the ambiguous licence proxy, log vehicle value, seats, log power-to-weight ratio, selected squared terms, and categorical indicators for policy type, business type, payment frequency, bonus score, fuel, vehicle brand, municipality type, and circulation area. Numerical values were median-imputed and clipped at calibration-period 0.5% and 99.5% quantiles before standardization. Rare brands were pooled. The L2 penalty was fixed at  $(10^{-4})$ , excluding the intercept, and models were estimated by L-BFGS-B. Calendar trend and policy status were excluded.

Evaluation used aggregate relative error, policy-level Poisson deviance, risk-decile observed-to-predicted ratios, and approximate aggregate process intervals. These intervals combine Poisson count variation with the existing

lognormal cost-dispersion proxy. They exclude parameter, feature, model, and distribution-shift uncertainty and are therefore labelled process-only.

### 3.4. ARCCRAFT architecture

An ARCCRAFT world contains a scenario class, regime, initial values for eleven state variables, and a twelve-period path. The current synthetic implementation allocates PCG64 substreams to `world`, `claims`, and `behaviour` using `SeedSequence.spawn`. World states are drawn around a central state and perturbed by regime and scenario-class intensity. Scenario classes are baseline, alternative, adverse, severe-plausible, and exploratory; reverse stress is produced by a separate local search. Because no external probability or divergence criterion calibrates the labels, “severe-plausible” denotes a configured class name rather than demonstrated economic plausibility.

The validator first applies hard bounds that trigger rejection and then soft bounds that clip accepted states and record repair. Simulation uses copies of validated state arrays. Premium, claims, and a payment-traceability score accumulate over twelve periods while behavioural noise updates state variables when transitions are active. A world fails first on a non-compensatory traceability threshold of 0.95 and otherwise on a combined-ratio threshold of 1.0. The Failure Atlas stores the world identifier, regime, class, failed gate, output values, master seed, and stable substream descriptors. It currently records terminal gate attribution, not the full first-failure trajectory or a causal mechanism.

### 3.5. Procedural ablations

Five configurations were executed for master seeds 20260825, 20260826, and 20260827 with 10,000 worlds per seed:

- full ARCCRAFT;
- no validator, passing raw initial states into the simulator;
- strict rejection, changing every repair into rejection;
- no transitions, holding validated initial states fixed over all twelve periods;
- shared stream, replacing the three named substreams with one sequential PCG64 generator.

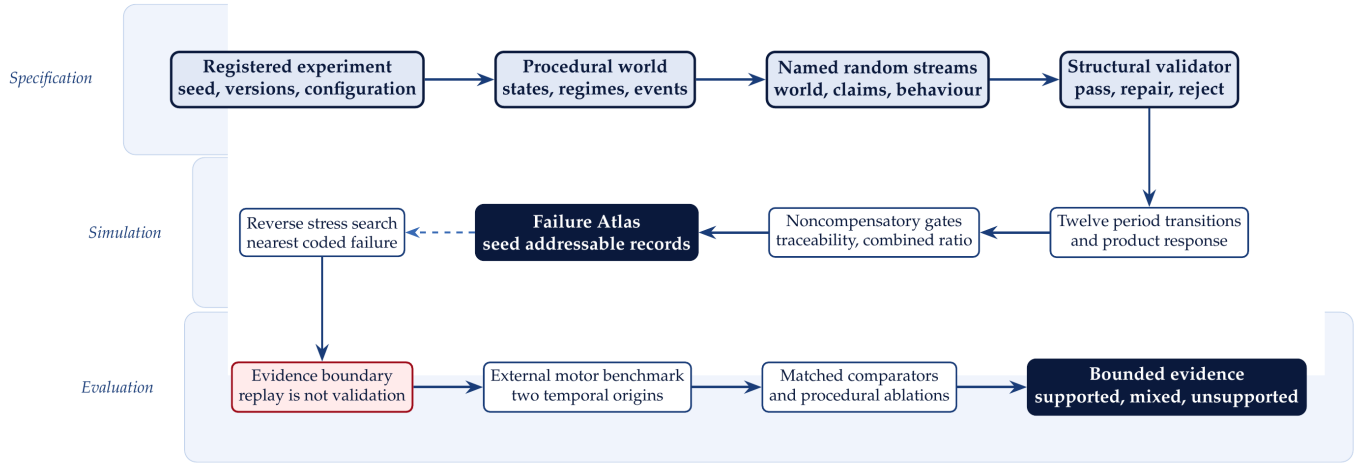
Every configuration was rerun and compared by a SHA-256 fingerprint of labels and numerical outputs. Stream isolation was tested by consuming 100 unused random draws after world generation and before simulation. The intervention should not affect claims or behaviour under separated streams but should affect a sequential shared stream. Outcomes included accepted sample size, repair and rejection counts, failure rate among accepted worlds, failure-or-rejection rate, gate-specific failures, mean combined ratio, mean traceability, rank correlation screening, and wall-clock runtime. Runtime is host-specific and descriptive.

## 4. Results and findings

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### 4.1. Data integrity and temporal shift

The database was mechanically consistent at the insured-year grain, but semantic and temporal



**Figure 1: ARCCRAFT procedural architecture and evidence boundaries.** The specification layer registers the experiment, generates procedural worlds, separates stochastic streams, and validates states before simulation. The simulation layer links transitions and noncompensatory gates to replayable Failure Atlas records and reverse stress search, while the evaluation layer keeps computational replay distinct from empirical validation and classifies the resulting evidence as supported, mixed, or unsupported.

risks were material. The mismatch between the licence-field definition and observed values prevented a substantive interpretation of that predictor. Zero-incurred claim rows ruled out an unmodified positive-only severity model. The rapid portfolio growth and composition shift made random row splitting inappropriate and justified temporal evaluation. These limitations were handled in modelling but cannot be removed without the source documentation and claim-level records.

## 4.2. External temporal benchmark

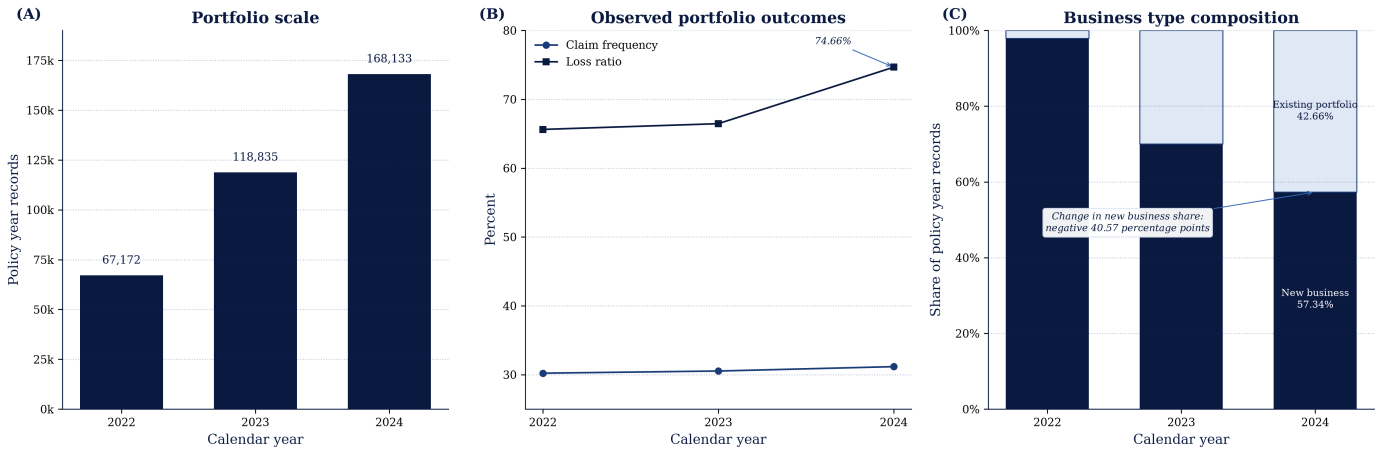
Both policy models converged at both origins. The 2023 design contained 54 encoded features and the 2024 design 63; fitted training totals reconciled approximately to observed totals.

Policy-level frequency improved held-out count deviance in both years and improved the 2023 aggregate count error slightly. Its 2024 aggregate count error was slightly worse than B0. B1

also achieved lower incurred deviance because differentiated count predictions change expected incurred cost even with pooled severity.

B2 did not demonstrate severity-model superiority. It reduced the absolute 2024 incurred error by 0.43 percentage points relative to B0, but increased the 2023 error from +2.70% to +9.58% and had higher incurred deviance than B1 in both years. This temporal instability is a negative result.

Calibration-by-decile also qualified the deviance improvement. In the lowest predicted count-risk decile, observed-to-predicted ratios were 1.31 in 2023 and 1.44 in 2024. Incurred-cost ratios in the lowest decile were approximately 1.68 and 1.93, respectively. The miss was therefore structured across risk groups rather than confined to aggregate sampling noise. All incurred totals fell outside their process-only intervals, and all three 2024 count forecasts missed their intervals. With only two origins, long-run interval coverage and tail calibration remain unidentifi-



**Figure 2: Motor portfolio scale, outcome dynamics, and composition shift, 2022 to 2024.** Panel A reports the growth from 67,172 to 168,133 policy year records. Panel B shows annual claim frequency and loss ratio, while Panel C shows the decline in the new business share from 97.91% to 57.34%, a 40.57 percentage point change that supports temporal rather than random evaluation. Source: public Spanish motor insurance portfolio, 2022 to 2024.

**Table 1: Temporal forecast performance across matched actuarial comparators.**

| Target | Model                                | Claim error | Count deviance | Count in 95% process interval | Incurred error | Incurred deviance | Incurred in 95% process interval |
|--------|--------------------------------------|-------------|----------------|-------------------------------|----------------|-------------------|----------------------------------|
| 2023   | B0 aggregate pooled                  | -1.03%      | 0.8481         | Yes                           | +2.70%         | 1200.08           | No                               |
| 2023   | B1 policy frequency, pooled severity | -0.86%      | 0.7871         | Yes                           | +2.87%         | 1150.35           | No                               |
| 2023   | B2 policy frequency, policy severity | -0.86%      | 0.7871         | Yes                           | +9.58%         | 1174.04           | No                               |
| 2024   | B0 aggregate pooled                  | -2.38%      | 0.9153         | No                            | -4.88%         | 1353.54           | No                               |
| 2024   | B1 policy frequency, pooled severity | -2.43%      | 0.8330         | No                            | -4.92%         | 1296.26           | No                               |
| 2024   | B2 policy frequency, policy severity | -2.43%      | 0.8330         | No                            | -4.44%         | 1302.42           | No                               |

able.

### 4.3. Synthetic engine and Failure Atlas

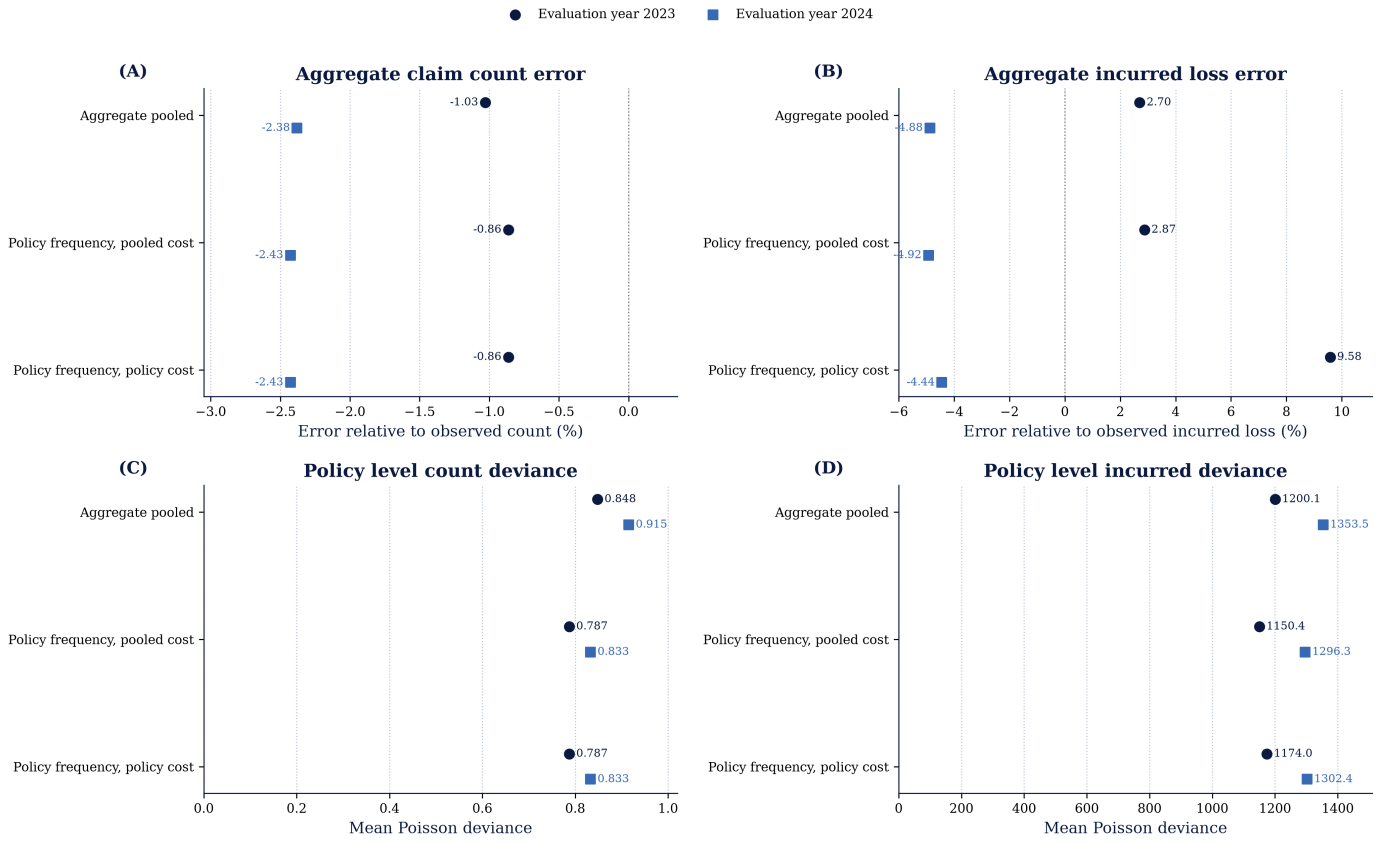
The separate aggregate sensitivity analysis produced a monotone 5 by 5 frequency and severity response surface under the implemented multiplicative shocks. At a joint frequency multiplier of 1.25 and severity multiplier of 1.25, the mean loss ratio was 1.10985, with a 95% process interval from 1.09131 to 1.12822. This verifies the configured response and identifies a technical failure region, but it does not establish that the joint shock is historically calibrated or economically plausible.

The canonical full run at seed 20260825 yielded 8,037 pass, 1,958 repair, and five reject

worlds. The Failure Atlas retained 3,588 traceability failures, one combined-ratio failure, and all five structural rejects. The reverse-stress search found a configured failure at standardized distance 1.0123, with traceability 0.9321 and combined ratio 0.4017. This is a local search result under the synthetic grammar, not proof of a globally minimal or economically plausible stress.

The imbalance between 3,588 traceability failures and one combined-ratio failure is central to interpretation. The present experiment mainly evaluates an operational threshold embedded in the simulator; it offers almost no statistical power for comparing actuarial failure detection. Mean combined ratios around 0.44 in the ablation runs are also far below the failure threshold.

### Matched temporal forecast performance



**Figure 3: Temporal forecast performance across matched actuarial comparators.** Panels A and B report aggregate errors relative to observed claim counts and incurred losses, where zero denotes perfect aggregate prediction. Panels C and D report policy level Poisson deviance, where smaller values indicate better predictive discrimination; policy level frequency improves deviance in both years, but the policy level cost model produces a pronounced 2023 aggregate error and never dominates the pooled cost comparator. Source: retrospective ARCCRAFT benchmark using the public Spanish motor insurance portfolio.

#### 4.4. Reproducibility, ablation, and cost

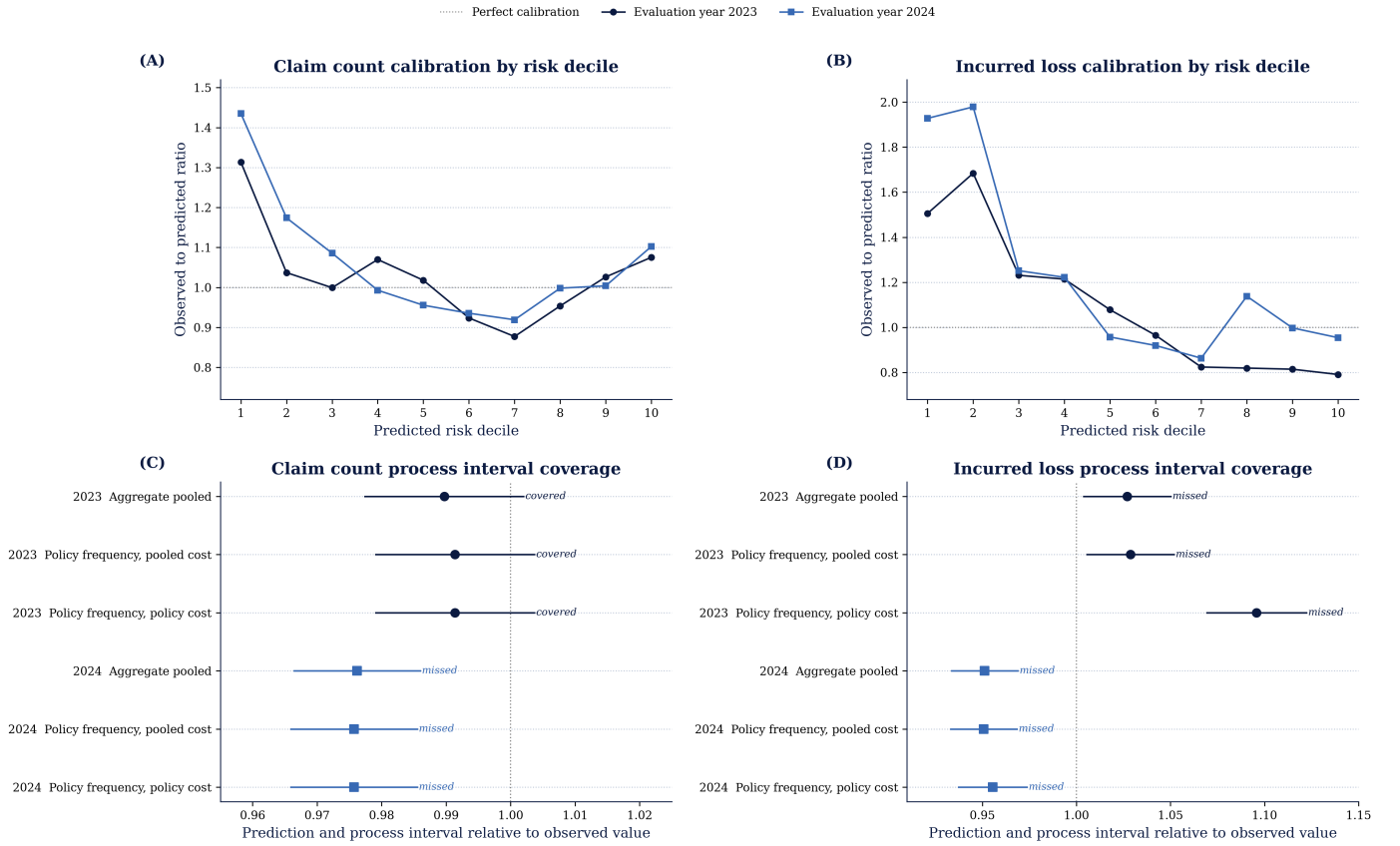
All 15 configuration-seed pairs replayed with identical fingerprints. Under the full configuration, consuming 100 unused world-stream draws before simulation left the fingerprint unchanged. Under the shared-stream configuration, the same intervention changed the fingerprint. H1 and H2 are therefore supported as local computational properties.

The full validator repaired an average of 1,986.3 worlds and rejected 4.7 per seed. Strict rejection accepted only the pass population and increased the total failure-or-rejection rate to

49.41%. This supports H3 only in the narrow sense that validator policy changes selection and reporting; no external criterion establishes that repair creates more plausible worlds.

Disabling transitions lowered the accepted-world failure rate by 0.25 percentage points relative to the full model. This supports H4 as a detectable computational effect, but the effect is small under the current grammar and three-seed design. The shared-stream failure-rate difference is not interpretable as an actuarial effect because sharing changes the random-number coupling. Its relevant result is the failed isolation test, not the lower observed rate.

### Calibration and process interval diagnostics

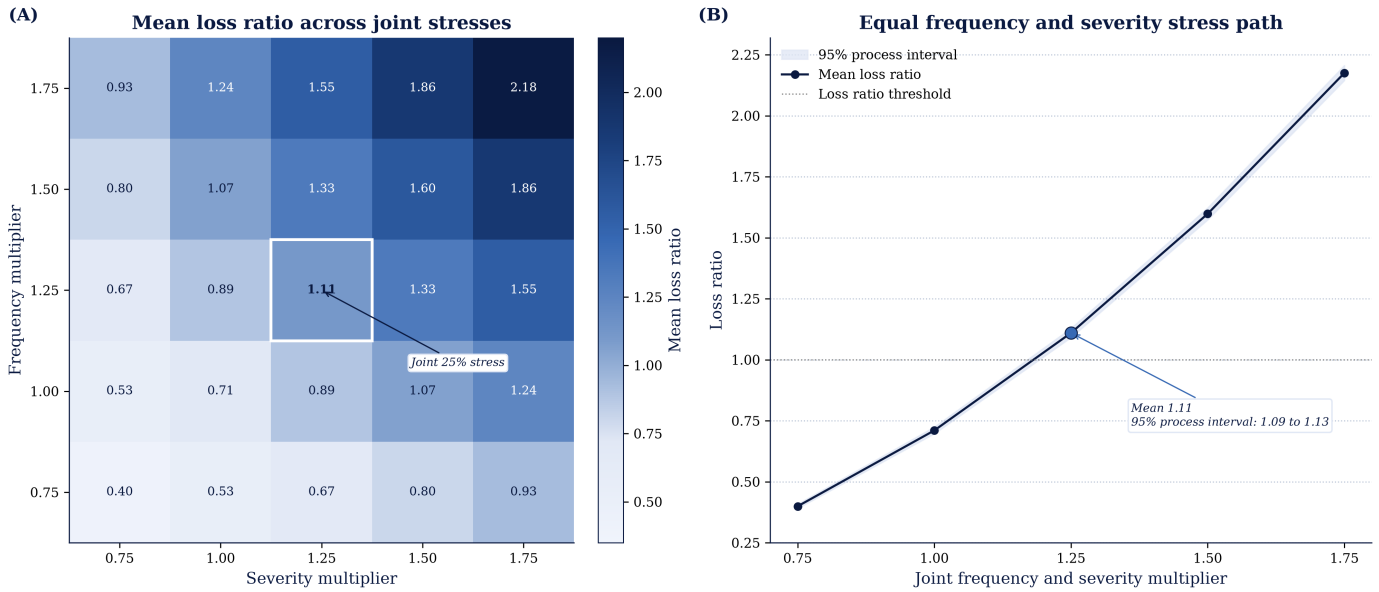


**Figure 4: Risk decile calibration and process interval coverage.** Panels A and B plot observed to predicted ratios by predicted risk decile, with 1.00 denoting perfect calibration; the lowest predicted risk groups are materially underpredicted, particularly for 2024 incurred loss. Panels C and D normalize predictions and their 95% process only intervals to the observed aggregate, showing count coverage in 2023, count misses in 2024, and incurred loss misses for every comparator and year. Source: retrospective ARCCRAFT benchmark using the public Spanish motor insurance portfolio.

Runtime differences were modest and noisy. The full pipeline averaged 3.59 seconds, while configuration means ranged from 3.21 to 4.74 seconds. Removing the validator was not faster in these wall-clock observations, indicating that incidental runtime variation and changed downstream calculation dominate the validator’s small direct cost. These timings should not be generalized across hosts.

To prevent qualitative interpretation from exceeding the measured evidence, the principal empirical claims are mapped to their numerical support below. Relative changes are descriptive because the benchmark has only two temporal

origins and the ablation has only three seeds per configuration.



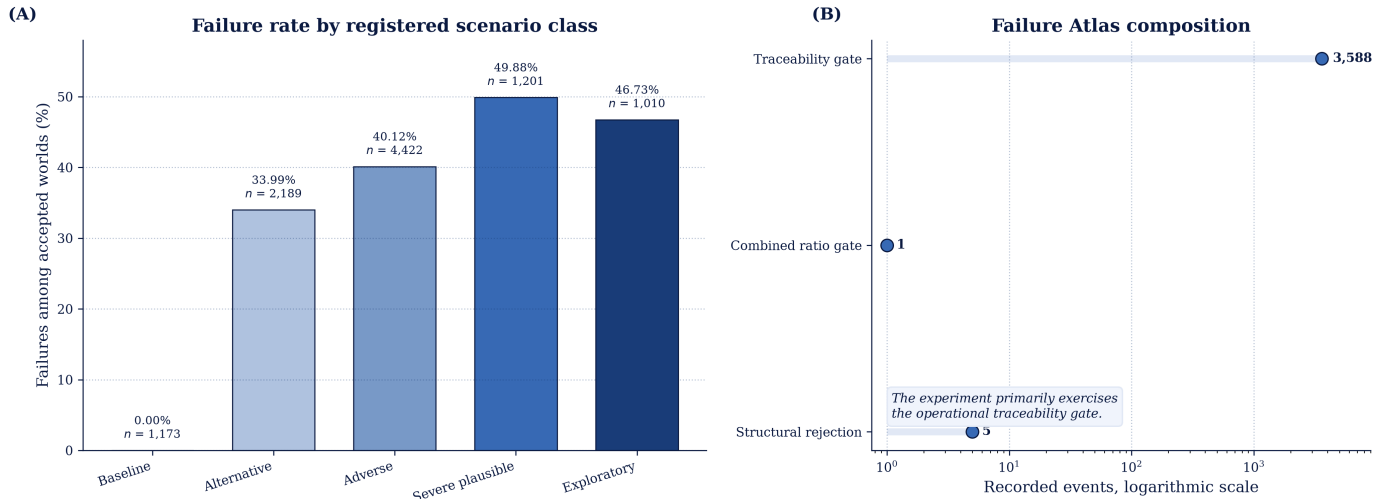
**Figure 5: Frequency and severity stress surface for the motor portfolio.** Panel A reports the mean simulated loss ratio for the 25 combinations of frequency and severity multipliers, with the outlined cell identifying the joint 25% stress. Panel B follows the equal multiplier path and shows the 95% process interval; the joint 1.25 multiplier produces a mean loss ratio of 1.11, above the technical threshold of 1.00. Source: registered ARCCRAFT stress simulations using the retrospective motor calibration.

**Table 2: Procedural ablation results across three registered seeds.**

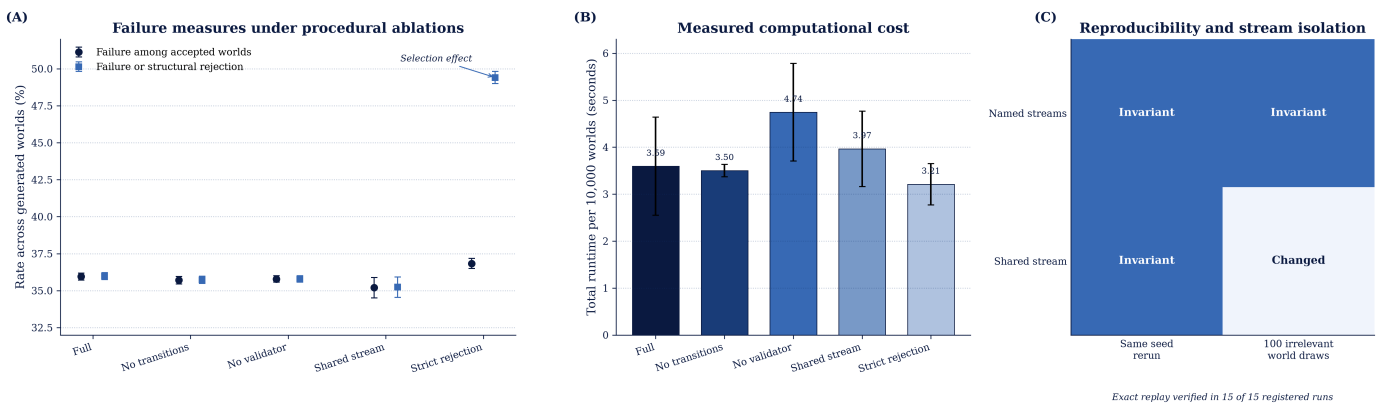
| Configuration  | Mean accepted-world failure rate | SD across three seeds | Mean failure-or-reject rate | Mean combined ratio | Mean runtime for 10,000 worlds |
|----------------|----------------------------------|-----------------------|-----------------------------|---------------------|--------------------------------|
| Full           | 35.95%                           | 0.24 pp               | 35.98%                      | 0.4436              | 3.59 s                         |
| No validator   | 35.79%                           | 0.22 pp               | 35.79%                      | 0.4436              | 4.74 s                         |
| Strict reject  | 36.83%                           | 0.34 pp               | 49.41%                      | 0.4363              | 3.21 s                         |
| No transitions | 35.70%                           | 0.25 pp               | 35.73%                      | 0.4432              | 3.50 s                         |
| Shared stream  | 35.20%                           | 0.69 pp               | 35.24%                      | 0.4429              | 3.97 s                         |

**Table 3: Claim-to-metric evidentiary audit.**

| Claim                                                           | Metric                                                                       | Observed result                                                                           | Evidential boundary                                                    |
|-----------------------------------------------------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Policy-level frequency improves discrimination                  | Relative reduction in held-out count deviance versus B0                      | 7.19% in 2023; 8.99% in 2024                                                              | Supports discrimination only, not aggregate calibration or superiority |
| Predictive intervals are inadequately calibrated                | Observed outcomes inside 95% process intervals                               | 3 of 12 target-model outcomes overall; 3 of 6 claim counts; 0 of 6 incurred totals        | Two origins cannot estimate stable long-run coverage                   |
| Registered runs replay exactly                                  | Identical output fingerprints                                                | 15 of 15 configuration-seed pairs, 100%                                                   | Local computational verification only                                  |
| Named streams isolate irrelevant world draws                    | Fingerprint response to 100 unused world-stream draws                        | Unchanged with named streams; changed with one shared stream                              | One registered intervention, not universal robustness                  |
| Validator policy changes the selected population                | Strict rejection versus full configuration                                   | 19.87% fewer accepted worlds; failure-or-rejection rate higher by 13.43 percentage points | Demonstrates selection and reporting effects, not greater plausibility |
| State transitions have a small observed effect                  | No-transition versus full accepted-world failure rate                        | Lower by 0.247 percentage points, 0.69% relative                                          | Descriptive result from three seeds                                    |
| The current Failure Atlas is dominated by the traceability gate | Share of canonical simulated failures attributed to traceability             | 3,588 of 3,589, 99.97%                                                                    | Does not establish causal attribution or diagnostic accuracy           |
| Runtime comparison is noisy                                     | Full-pipeline runtime mean, standard deviation, and coefficient of variation | 3.59 s; 1.05 s; 29.09%                                                                    | Host-specific descriptive timing                                       |



**Figure 6: Scenario class failure rates and Failure Atlas composition.** Panel A reports failures among accepted worlds for each registered scenario class in the canonical 10,000 world run, from 0.00% under Baseline to 49.88% under Severe plausible conditions; Exploratory worlds remain analytically separate. Panel B reports atlas events on a logarithmic axis and shows that 3,588 traceability failures dominate one combined ratio failure and five structural rejections, so the experiment primarily tests the operational gate rather than actuarial insolvency detection. Source: canonical ARCCRAFT run, registered seed 20260825.



**Figure 7: Procedural ablations, computational cost, and stream isolation.** Panel A compares failures among accepted worlds with the broader failure or rejection measure across three registered runs per configuration; strict rejection creates the largest divergence because repaired worlds become structural rejections. Panel B reports mean runtime with one standard deviation, while Panel C shows exact replay in all 15 runs and confirms that named streams preserve claims and behavioural outputs after 100 irrelevant world draws, whereas a shared stream does not. Source: ARCCRAFT ablation and stream isolation experiments at registered seeds 20260825, 20260826, and 20260827.

## 5. Discussion

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The study supports a narrower contribution than the phrase “validated framework” would imply. ARCCRAFT provides a locally reproducible computational architecture that makes seeds, validation decisions, scenario classes, and failure gates explicit. All 15 registered configuration-seed pairs replayed with identical fingerprints, and the 100-draw intervention left the named-stream fingerprint unchanged while changing the shared-stream fingerprint. These metrics establish local replay and component isolation. They

do not measure operational usefulness through a user study or deployment evaluation.

The empirical benchmark gives mixed evidence. Policy-level frequency reduces held-out count deviance by 7.19% in 2023 and 8.99% in 2024 relative to the aggregate comparator, but it does not uniformly improve aggregate errors or calibration. Only 3 of 12 reported target-model outcomes fall inside their 95% process intervals, including none of the six incurred totals. The policy-level cost model fails the intended superiority test because its 2023 incurred error increases from +2.70% to +9.58%, and the lowest incurred-risk decile has an observed-to-predicted ratio of 1.68. The result illustrates why aggregate error, individual deviance, group calibration, and interval coverage must remain separate.

The procedural ablations also delimit the architectural claims. Strict rejection reduces the accepted population by 19.87% relative to the full configuration and increases the failure-or-rejection rate by 13.43 percentage points, while the accepted-world failure rate rises by only 0.88 percentage points. The validator is therefore part of the data-generating process, not a neutral cleaning step. Yet the experiment provides no statistical or economic plausibility criterion for choosing repair over rejection. Disabling transitions lowers the accepted-world failure rate by 0.247 percentage points, or 0.69% relative to the full configuration. This is a small descriptive effect under the present parameterization and three-seed design, not evidence of general transition importance.

The canonical Failure Atlas records 3,594 events that reconcile to 3,588 traceability failures, one combined-ratio failure, and five structural rejections. Traceability represents 99.97% of sim-

ulated failures, showing that the current experiment primarily audits one operational gate. This supports event recording and replay, but not diagnostic accuracy. The records do not contain an empirically observed failure label, implanted causal mechanism, or full first-failure time. Controlled fault-injection experiments should therefore be added before reporting precision, recall, detection delay, attribution rank, or causal identification.

For management and model-risk governance, the measured contribution is procedural: registered runs replayed exactly, stream isolation passed the specified intervention, and validator policy changed the selected population by a measured amount. Whether these properties improve model challenge in practice remains untested. Operational value requires user studies, deployment evidence, or governance-case evaluation that is not available here.

### 5.1. Limitations

The motor data cover only three annual periods, the final year was already inspected, and the original raw motor file is absent even though its recorded hash is available. Policy-year incurred totals do not identify individual claim severity. Process intervals omit parameter and model uncertainty. The full synthetic grammar is not calibrated to the motor portfolio, and scenario class probabilities, transitions, dependencies, and thresholds are assumptions. Only three ablation seeds were timed on one host. Combined-ratio failures are too rare for useful comparison. No external operational labels or independent portfolio are available. These constraints prevent claims of general predictive validation, stable tail coverage, causal attribution, or external validity.

## 6. Conclusion and recommendations

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ARCCRAFT is best characterized by the present evidence as a proposed reproducible audit architecture for procedural actuarial stress testing. In the inspected local environment, 15 of 15 registered runs replay exactly, the named-stream fingerprint is invariant to 100 irrelevant world draws, strict rejection reduces accepted worlds by 19.87%, and disabling transitions changes the accepted-world failure rate by 0.247 percentage points. These findings support local computational verification and quantifiable architectural effects.

The evidence does not support overall predictive superiority. Policy-level frequency modelling improves held-out count deviance but remains temporally miscalibrated, while policy-level severity is unstable. The 2024 outcome falls outside all reported process-only intervals, and the synthetic experiment is dominated by its traceability threshold. The paper therefore rejects complete validation and treats the empirical results as retrospective preliminary evidence.

Four next steps are required. First, evaluate a prospectively untouched year or independent portfolio. Second, obtain claim-level payments, expenses, commissions, and operational traceability outcomes so that loss tails and failure gates can be validated empirically. Third, calibrate scenario dynamics and plausibility criteria rather than relying on labels. Fourth, conduct registered fault-injection experiments and compare diagnostic precision, recall, delay, and attribution rank against conventional simulation and sensitivity methods. Until those steps are completed, ARCCRAFT's strongest demonstrated contribution

is reproducible procedural integration, not established actuarial superiority.

## Data, code, and reproducibility statement

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All reported calculations are stored in the local research package. The policy benchmark is executed in `17_DATA_QUALITY_POLICY_BENCHMARK.ipynb`; machine-readable outputs are files 18-24. The procedural ablation is implemented in `run_arccraft_ablation.py`; outputs are files 25-28. The current repository base commit is `b3df670`, with documented research-related working-tree changes. The normalized SQLite database is included in the repository, but the original motor source file is not bundled. No personal identities are reported in this paper.

An integrated public demonstrator is accessible through Figure 8. On 25 September 2026, the deployment returned API version 2.1.0 and successfully executed the synthetic ARCCRAFT endpoint. Its compiled interface logic, data module, API client, and styling matched the corresponding local assets byte for byte. The startup-theme document and parts of the public backend differed from the current local working tree, including the reverse-stress result and stochastic-stream metadata. The demonstrator is therefore suitable for interface inspection and live workflow demonstration, but it is not the authoritative computational artifact for the numerical results reported in this paper.



**Figure 8: Public ARCCRAFT demonstrator access.** The QR code opens the ARCCRAFT module within the integrated MAPTA, COMPASS, and ARCCRAFT demonstrator. The public deployment exposes API version 2.1.0 and executes the synthetic simulation endpoint. It is provided for interface inspection rather than as the authoritative source of the paper’s numerical results because parts of the deployed backend differ from the current local working tree.

## References

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- [1] José Garrido, Christian Genest, and Juliana Schulz. [Generalized linear models for dependent frequency and severity of insurance claims](#). *Insurance: Mathematics and Economics*, 70:205–215, 2016. doi: 10.1016/j.insmatheco.2016.06.006.
- [2] Peng Shi, Xiangnan Feng, and Anastasia Ivantsova. Dependent frequency-severity modeling of insurance claims. *Insurance: Mathematics and Economics*, 64:417–428, 2015.
- [3] Edward W. Frees, Gee Y. Lee, and Lu Yang. [Multivariate frequency-severity regression models in insurance](#). *Risks*, 4(1):Article 4, 2016. doi: 10.3390/risks4010004.
- [4] Gee Y. Lee and Peng Shi. [A dependent frequency-severity approach to modeling longitudinal insurance claims](#). *Insurance: Mathematics and Economics*, 87:115–129, 2019. doi: 10.1016/j.insmatheco.2019.04.004.
- [5] Himchan Jeong and Emiliano A. Valdez. [Predictive compound risk models with dependence](#). *Insurance: Mathematics and Economics*, 94:182–195, 2020. doi: 10.1016/j.insmatheco.2020.07.011.
- [6] Guangyuan Gao and Jackie Li. [Dependence modeling of frequency-severity of insurance claims using waiting time](#). *Insurance: Mathematics and Economics*, 109:29–51, 2023. doi: 10.1016/j.insmatheco.2022.12.006.
- [7] Nadja Klein, Michel Denuit, Stefan Lang, and Thomas Kneib. [Nonlife ratemaking and risk management with Bayesian generalized additive models for location, scale, and shape](#). *Insurance: Mathematics and Economics*, 55:225–249, 2014. doi: 10.1016/j.insmatheco.2014.02.001.
- [8] European Insurance and Occupational Pensions Authority. [Methodological principles of insurance stress testing](#), 2020.
- [9] Thomas Breuer, Martin Jandacka, Klaus Rheinberger, and Martin Summer. How to find plausible, severe, and useful stress scenarios. *International Journal of Central Banking*, 5(3): 205–224, 2009.
- [10] Thomas Breuer and Imre Csiszar. [Systematic stress tests with entropic plausibility constraints](#). *Journal of Banking & Finance*, 37(5):1552–1559, 2013. doi: 10.1016/j.jbankfin.2012.04.013.
- [11] Paul Glasserman, Chulmin Kang, and Wanmo Kang. [Stress scenario selection by empirical likelihood](#). *Quantitative Finance*, 15(1):25–41, 2015. doi: 10.1080/14697688.2014.926019.
- [12] Mark D. Flood and George G. Korenko. [Systematic scenario selection: Stress testing and the nature of uncertainty](#). *Quantitative Finance*, 15(1):43–59, 2015. doi: 10.1080/14697688.2014.926018.

- [13] Silvana M. Pesenti, Pietro Millossovich, and Andreas Tsanakas. [Reverse sensitivity testing: What does it take to break the model?](#) *European Journal of Operational Research*, 274(2):654–670, 2019. doi: 10.1016/j.ejor.2018.10.003.
- [14] Elias Kroell, Silvana M. Pesenti, and Sebastian Jaimungal. [Stressing dynamic loss models.](#) *Insurance: Mathematics and Economics*, 114:56–78, 2024. doi: 10.1016/j.insmatheco.2023.11.002.
- [15] Max D. Morris. [Factorial sampling plans for preliminary computational experiments.](#) *Technometrics*, 33(2):161–174, 1991. doi: 10.1080/00401706.1991.10484804.
- [16] Andrea Saltelli, Marco Ratto, Terry Andres, Francesca Campolongo, Jessica Cariboni, Debora Gatelli, Michaela Saisana, and Stefano Tarantola. [Global sensitivity analysis: The primer.](#) Wiley, 2008. doi: 10.1002/9780470725184.
- [17] Andrea Saltelli, Paola Annoni, Ivano Azzini, Francesca Campolongo, Marco Ratto, and Stefano Tarantola. [Variance based sensitivity analysis of model output: Design and estimator for the total sensitivity index.](#) *Computer Physics Communications*, 181(2):259–270, 2010. doi: 10.1016/j.cpc.2009.09.018.
- [18] Andrea Gabrielli and Mario V. Wuthrich. [An individual claims history simulation machine.](#) *Risks*, 6(2):Article 29, 2018. doi: 10.3390/risks6020029.
- [19] Benjamin Avanzi, Greg Taylor, Melantha Wang, and Bernard Wong. [SynthETIC: An individual insurance claim simulator with feature control.](#) *Insurance: Mathematics and Economics*, 100: 296–308, 2021. doi: 10.1016/j.insmatheco.2021.06.004.
- [20] Benjamin Avanzi, Greg Taylor, and Melantha Wang. [SPLICE: A synthetic paid loss and incurred cost experience simulator.](#) *Annals of Actuarial Science*, 17(1):7–35, 2023. doi: 10.1017/S1748499522000057.
- [21] M.-O. Wolf, B. N. Meyer, P. Mahler, and M. Diehl. [openIRM: Publicly accessible internal risk model of an artificial life insurer for analyzing and benchmarking actuarial methods in the Solvency II setting.](#) *European Actuarial Journal*, 16:225–282, 2026. doi: 10.1007/s13385-025-00435-6.
- [22] Roger D. Peng. [Reproducible research in computational science.](#) *Science*, 334(6060):1226–1227, 2011. doi: 10.1126/science.1213847.
- [23] Victoria Stodden. [Reproducing statistical results.](#) *Annual Review of Statistics and Its Application*, 2:1–19, 2015. doi: 10.1146/annurev-statistics-010814-020127.
- [24] Victoria Stodden, Marcia McNutt, David H. Bailey, Ewa Deelman, Yolanda Gil, Brooks Hanson, Michael A. Heroux, John P. A. Ioannidis, and Michela Taufer. [Enhancing reproducibility for computational methods.](#) *Science*, 354(6317):1240–1241, 2016. doi: 10.1126/science.aah6168.

- [25] Pierre L'Ecuyer, Richard Simard, E. Jack Chen, and W. David Kelton. [An object-oriented random-number package with many long streams and substreams](#). *Operations Research*, 50(6): 1073–1075, 2002. doi: 10.1287/opre.50.6.1073.358.
- [26] Robert G. Sargent. [Verification and validation of simulation models](#). *Journal of Simulation*, 7(1): 12–24, 2013. doi: 10.1057/jos.2012.20.
- [27] Jack P. C. Kleijnen and Robert G. Sargent. [A methodology for fitting and validating metamodels in simulation](#). *European Journal of Operational Research*, 120(1):14–29, 2000. doi: 10.1016/S0377-2217(98)00392-0.
- [28] Actuarial Standards Board. [Actuarial Standard of Practice No. 56: Modeling](#), 2020.
- [29] Armen Der Kiureghian and Ove Ditlevsen. [Aleatory or epistemic? Does it matter?](#) *Structural Safety*, 31(2):105–112, 2009. doi: 10.1016/j.strusafe.2008.06.020.
- [30] Marc C. Kennedy and Anthony O'Hagan. [Bayesian calibration of computer models](#). *Journal of the Royal Statistical Society: Series B*, 63(3):425–464, 2001. doi: 10.1111/1467-9868.00294.
- [31] Pauline Barrieu and Giacomo Scandolo. [Assessing financial model risk](#). *European Journal of Operational Research*, 242(2):546–556, 2015. doi: 10.1016/j.ejor.2014.10.032.
- [32] Tilmann Gneiting and Adrian E. Raftery. [Strictly proper scoring rules, prediction, and estimation](#). *Journal of the American Statistical Association*, 102(477):359–378, 2007. doi: 10.1198/016214506000001437.
- [33] Tilmann Gneiting, Fadoua Balabdaoui, and Adrian E. Raftery. [Probabilistic forecasts, calibration and sharpness](#). *Journal of the Royal Statistical Society: Series B*, 69(2):243–268, 2007. doi: 10.1111/j.1467-9868.2007.00587.x.
- [34] Claudia Czado, Tilmann Gneiting, and Leonhard Held. [Predictive model assessment for count data](#). *Biometrics*, 65(4):1254–1261, 2009. doi: 10.1111/j.1541-0420.2009.01191.x.
- [35] Paul H. Kupiec. [Techniques for verifying the accuracy of risk measurement models](#). *Journal of Derivatives*, 3(2):73–84, 1995. doi: 10.3905/jod.1995.407942.
- [36] Peter F. Christoffersen. [Evaluating interval forecasts](#). *International Economic Review*, 39(4): 841–862, 1998. doi: 10.2307/2527341.
- [37] Natalia Nolde and Johanna F. Ziegel. [Elicitability and backtesting: Perspectives for banking regulation](#). *Annals of Applied Statistics*, 11(4):1833–1874, 2017. doi: 10.1214/17-AOAS1041.
- [38] Maxime Cauchois, Suyash Gupta, Alnur Ali, and John C. Duchi. [Robust validation: Confident predictions even when distributions shift](#). *Journal of the American Statistical Association*, 119(548): 3033–3044, 2024. doi: 10.1080/01621459.2023.2298037.

- [39] Carlo Acerbi and Dirk Tasche. [On the coherence of expected shortfall](#). *Journal of Banking & Finance*, 26(7):1487–1503, 2002. doi: 10.1016/S0378-4266(02)00283-2.